

DSST389: Advanced Data Science — Taylor Arnold — Spring 2026

Website: <https://taylor-arnold.github.io/courses/dsst389-s26>

Topics: Advanced approaches to the analysis of data. Focuses on putting together larger, complete projects that combine quantitative and qualitative analyses.

Format: The semester is broken into four units, with an assessment at the end of each. Class meetings are hands-on and interactive. Please bring a computer, pencil, and something to write on to each class. Class materials can be found on the course website.

Software: We will be programming in the open-source Python programming language. Classwork can be run through Google's Colab environment in a web browser using a University email login. Each student will also have a personal shared Box folder that will be used to submit and return materials and grades. Other hardware and software may be used for some data collection tasks; all of these will be provided or freely accessible.

Notebooks: We will work through Python notebooks during class. Completed notebooks should be uploaded to Box before the start of the following class. It is not permitted to use generative AI or the posted solutions during class unless specifically asked to. Any resources are permitted for finishing the notebooks on your own.

Class Engagement Grade: The class engagement grade is graded on a 100-point scale, but everyone begins with 120 points. Five points are deducted from this total for each missed class meeting and for each missed uploaded notebook. This system effectively builds in approximately five missed classes or assignments while still allowing you to earn full credit. It is intended to accommodate routine illnesses and other unavoidable circumstances.

External Speakers: As part of this course, you are required to attend at least two of the three external speaker presentations listed on the course website. These presentations are a required component of the course but are not associated with a numerical grade. You must contact the instructor as soon as possible if you are unable to attend two or more of the presentations, so that appropriate arrangements can be discussed.

Assessments: There are four assessments in this course. The first three assessments are in-class midterm exams, given on the dates indicated on the course website. The fourth assessment is an open-book, open-notes written assignment due on the last day of class. A list of topics for each of the in-class exams will be posted on the course website. There is no final exam.

Getting Help: We will usually have time in class to answer questions about the course material. If questions arise outside of class, please feel free to send these by email. I am happy to schedule office hours for any extended questions or personal concerns. Because I know everyone has a busy schedule, I offer office hours primarily by appointment.

Final Grades: To get a final grade for the semester, the four assessment grades are averaged. The *minimum* score of the exam average and the engagement grade is computed. A letter grade is assigned as follows: A (93–100), A- (90–92), B+ (87–89), B (83–86), B- (80–82), C+ (77–79), C (73–76), C- (70–72), and F (0–69).